

Exploring Art in Medicine through Knowledge Graphs

Thesis Draft University of Amsterdam

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Abstract

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1. Introduction

1.1 Digital Exploration of Cultural Collections

Cultural heritage collections are increasingly accessible online, however accessibility alone does not equal meaningful exploration. Large digitized repositories have been built by museums, archives, and digital libraries but the dominant interaction model, a search box and a list of results, still treats the collection as a flat inventory rather than a web of connected knowledge. Several researchers have argued that effective access to cultural heritage requires moving beyond simple retrieval and toward systems that support structured, multidimensional exploration of collections (Shiri et al., 2015; Ardissono et al., 2016).

The Bagatelle collection examined in this project is a specific case of this broader challenge. It contains roughly 600 artworks relating to art in medicine and works depicting medical conditions, clinical settings, anatomical studies, and historical themes of care and disease. This is not a collection that responds well to keyword search. A query for "plague" will not return artworks that depict the social isolation of the sick, or the symbolic use of the crow, without explicit metadata pointing to those concepts. The collection's richness lives in the relationships between works, not in the labels assigned to them.

This gap between what a collection contains and what a user can discover through it is the motivating problem of this thesis.

1.2 Limitations of Keyword-Based and Metadata-Only Retrieval

The limitations of keyword-based search in cultural heritage contexts are well documented. Standard information retrieval models assume that a user can formulate a query that matches terms in the collection's metadata. But cultural and artistic collections often don't conform to this: their meaning is layered, historically contingent, visually encoded, and medically or symbolically specific in ways that metadata fields rarely capture (Shiri et al., 2015).

ArtSeek, a recent multimodal framework for artwork analysis, identifies this same gap as a central design challenge: most digitized collections lack structured links to external knowledge bases like Wikidata or Wikipedia, meaning that conventional entity-based search cannot function at all (Fanelli et al., 2025). A system that only indexes what is explicitly labeled will logically fail to surface what is implicitly meaningful.

The Bagatelle system prior to this project already acknowledged this limitation. Its starting architecture included LLM-generated textual descriptions of artworks and multimodal embeddings, a deliberate step away from raw metadata retrieval and toward semantic representation. The goal of this thesis is to build on that foundation, not just to improve retrieval, but to change the mode of interaction from search to exploration.

1.3 Multimodal and Semantic Representations of Artworks

A significant body of recent work has established that artworks are best represented by a combination of visual and textual signals instead of by a single feature space. Castellano et al. (2022) developed ArtGraph, a knowledge graph linking artworks to artists, styles, and periods. Their research

demonstrated that combining visual embeddings from a Vision Transformer with graph-structural embeddings from the knowledge graph outperforms purely visual approaches in style and genre classification (Castellano et al., 2022). The key insight is that semantic context (what an artwork is about, who made it, in what tradition) is information that visual features alone cannot encode.

ArtSeek extends this further by integrating retrieval-augmented generation: a multimodal large language model retrieves contextual knowledge fragments about an input artwork and uses those to reason about visual motifs, historical context, and artistic meaning (Fanelli et al., 2025). This approach is important for the Bagatelle context because it explicitly targets collections that lack external links, exactly the situation faced here.

In this project, the Bagatelle server already provides both image-based CLIP embeddings and LLM-generated textual descriptions from three models (GPT-4o, Claude, and GPT-5). These form the multimodal foundation on which the graph-based exploration layer is built. The choice of which representation to use: visual, textual, or combined, becomes a research question itself: different representations surface different kinds of relationships between artworks.

1.4 Knowledge Graphs as an Exploration Structure

A knowledge graph represents entities as nodes and relationships as typed, directed edges. This structure does something that a flat vector database cannot: it makes the nature of connections explicit. Rather than just knowing that two artworks are close in embedding space, a knowledge graph can record that they share a depicted disease, a historical period, an artistic technique, or a symbolic motif.

Prior work on knowledge graphs in cultural heritage shows both the promise and the complexity of this approach. The Linked Stage Graph project (FIZ Karlsruhe) built a knowledge graph from roughly 7,000 historical theatre photographs and their metadata, enabling exploration via linked data and graph-based navigation. A key challenge they identified is data sparsity: even a large collection may lack the structured metadata needed to construct a dense, navigable graph (Tietz et al., 2020). This is a problem directly relevant to the Bagatelle collection, where explicit medical and artistic metadata is limited.

At a larger scale, work on the Europeana database has demonstrated that graph-based clustering of art objects, using topic modeling to surface semantic groupings, can reveal associations that no individual metadata field would expose (Kouretsis et al., 2022). For the purposes of this project, the graph is not constructed from pre-existing metadata alone, but generated dynamically from embedding similarity and LLM-derived semantic relationships, making the schema itself a design decision.

Huang et al. (2023) address the challenge of fragmented cultural heritage data directly: they propose a joint entity-relation extraction model to build knowledge graphs from sparse, heterogeneous museum records, showing that knowledge completion techniques can improve both the density and usefulness of such graphs. This provides methodological context for the design choices made in this thesis, even though the approach taken here uses embedding-based rather than ontology-based relationship construction.

1.5 Interpretability and Exploratory Search

A recurring theme across this literature is that retrieval and exploration are different goals, and that systems designed only for retrieval often fail at exploration. Retrieval optimizes for returning the most relevant result to a given query. Exploration optimizes for discovering connections that the user did not know to look for.

The Linked Stage Graph work explicitly argues that for a broad range of users to engage meaningfully with cultural heritage data, multiple dimensions of the collection must be navigable simultaneously (Tietz et al., 2020). ArtSeek frames this as an interpretability challenge: a system should not only retrieve related artworks but enable the user to understand why they are related (Fanelli et al., 2025). This distinction between showing a result and explaining a connection will shape the design of the system built in this thesis.

In the current implementation, each graph edge carries both a numerical similarity score and an LLM-generated explanation of the semantic relationship between two artworks. This makes the graph inspectable: a user navigating from a surgical scene in the nineteenth century to a plague allegory from the Renaissance can read why the system considers them connected, rather than simply trusting that it does. This aligns with the growing interest in explainable AI for cultural heritage applications, where transparency in retrieval rationale directly affects user trust and educational value.

1.6 Comparing Relation Strategies

A central contribution of this project is not just building a graph-based exploration system, but building one that supports multiple, configurable strategies for generating graph connections. Prior work has mostly treated the relationship between artworks as a fixed property, determined by metadata, by a single embedding model, or by a pre-defined ontology. This project takes the position that the right strategy depends on the exploration goal.

The ArtSeek framework provides the closest case in point for this kind of multimodal, retrieval-oriented flexibility (Fanelli et al., 2025). Though it operates at the level of individual queries rather than graph construction. The graph embedding literature offers a broader technical basis: different embedding strategies, from shallow methods like node2vec to graph neural networks, capture different structural properties of the data (Xu, 2020).

In this project, five graph generation strategies are designed and compared: text-based (using LLM-generated descriptions), image-based (using CLIP visual embeddings), combined (fusing both), query-targeted (anchoring expansion to the user's original search intent), and LLM-confirmed (using the language model to judge the validity of candidate edges). The evaluation design draws on the observation that user preference is often the most meaningful measure of exploration quality.

1.7 Thesis Structure

This thesis proceeds as follows. Chapter 2 provides background on the relevant literature and positions this project within it. Chapter 3 describes the design of the extended Bagatelle system, including the graph schema, relation strategy architecture, and interface design. Chapter 4 covers the technical implementation. Chapter 5 presents the evaluation of the five graph strategies. Chapter 6 discusses findings, limitations, and future directions. Chapter 7 concludes.

2. Background and Related Work

This chapter will situate the project within relevant prior work across four areas: cultural heritage exploration systems, semantic and multimodal representations of artworks, knowledge graph construction in the cultural domain, and evaluation of exploratory search.

2.1 Cultural Heritage Digital Libraries and Exploration Systems

Overview of the shift from static digital libraries to interactive exploration systems. Discussion of key systems including Europeana, and challenges of scale, metadata sparsity, and user diversity.

2.2 Keyword-Based Retrieval and Its Limits in Art Collections

Detailed treatment of why standard IR models fail for visually and symbolically dense art collections. The missing link between indexable metadata and embodied visual meaning.

2.3 Multimodal Representations: CLIP, LLMs, and Art

Technical background on CLIP (image-text alignment), its strengths for cross-modal retrieval and weaknesses for pure text similarity. LLM-generated descriptions as an alternative textual signal. MiniLM as a text similarity model. Discussion of the Bagatelle server's existing embedding pipeline as the starting point.

2.4 Knowledge Graphs in Cultural Heritage

Overview of knowledge graph construction approaches: metadata-driven, ontology-based, clustering-based, and deep-learning-assisted.

2.5 Graph-Based Exploration Interfaces

How graph structures can be exposed to users. Layered layouts, node and edge labeling, interactive expansion. Tension between exploration breadth and comprehensibility.

2.6 Evaluation of Exploratory and Semantic Search

Challenges of evaluating systems where there is no single correct answer. Side-by-side preference comparison, edge relevance rating, and expert judgment as evaluation approaches.

3. System Design

This chapter describes the overall architecture of the extended Bagatelle system and the design decisions behind it.

3.1 Starting Point: The Bagatelle Server

Brief description of the existing system: Flask backend, Qdrant vector database, CLIP and MiniLM embeddings, LLM-generated HTML descriptions (GPT-4o, Claude, GPT-5), gallery interface. What was working and what was not before the project began.

3.2 Design Goals and Constraints

The core design goal: turn a search-results interface into a graph-based exploration tool. Key constraints: stay compatible with the existing Qdrant infrastructure, avoid introducing a separate graph database at this stage, keep the prototype usable. Design decision to build graph state in the JavaScript frontend and use Qdrant as the semantic neighbour engine.

3.3 Graph Schema

How artworks are represented as nodes and relationships as edges. What information is stored per node (image path, title, category, LLM description, embedding). What information is stored per edge (similarity score, source modality, LLM-generated explanation). Why a static graph database (e.g. Neo4j) was not used at this stage.

3.4 Graph Generation Strategies

Overview of the five strategies and the motivation for each:

- **3.4.1 Text-Based Graph** — Semantic similarity between LLM-generated descriptions using MiniLM embeddings. Comparison across GPT-4o, Claude, and GPT-5 descriptions.
- **3.4.2 Image-Based Graph** — Visual similarity via CLIP-L14 image embeddings from the supervisor-provided Qdrant snapshot.
- **3.4.3 Combined Graph** — Fused scoring from text and image retrieval. Weighted combination; slider interface.
- **3.4.4 Query-Targeted Graph** — Graph expansion anchored to the user's original search query, not just the clicked artwork's description.
- **3.4.5 LLM-Confirmed Graph** — Candidate edges generated by embedding similarity, then validated and explained by a language model.

3.5 Exclusion and Novelty in Graph Expansion

Design of the `exclude_paths` mechanism that prevents the graph from looping over already-seen artworks. Rationale: search relevance and exploration novelty are distinct goals, and a pure nearest-neighbour strategy causes stalemates in a small collection.

3.6 Interface Design

The gallery interface, clickable artworks, the graph view (layered left-to-right layout, visible edge scores, hoverable explanations), related artwork cards with expandable descriptions, and the clear/reset graph function. Design principle: every connection should be inspectable, not just visible.

4. Implementation

This chapter documents the technical implementation of the system, covering backend development, vector database management, and frontend graph rendering.

4.1 Backend Architecture

Flask application structure. Key routes: /retrieve (initial search), /related (graph expansion). Request/response formats. API versioning considerations.

4.2 Vector Database: Qdrant Setup and Schema

Collection structure, named vectors, payload indexing. Issues encountered: model mismatch between CLIP and MiniLM, schema errors, client API version changes (client.search → client.query_points). Data pipeline: HTML cleaning with BeautifulSoup, embedding generation, payload construction with title priority logic (CSV metadata → HTML title → LLM-derived → filename fallback).

4.3 Embedding Pipeline

Build scripts for text collections (build_text_collection.py). Image path resolution and extension handling (.jpg, .jpeg, .png, .webp, .jif, .gif, .avif). Missing image handling: non-fatal warnings, skip with logging.

4.4 LLM Integration

Claude API integration for edge explanation generation. Prompt design for relation explanation. Extension to LLM-confirmed mode: the model returns a relevance judgment, relation type, and explanation.

4.5 Frontend Graph Engine

JavaScript graph state: graphNodes (Map), graphEdges (Map), graphRoot. Layered depth-based layout algorithm. SVG edge rendering with invisible hitboxes for hover. Node labels from title payload. Edge score labels. Exclusion path tracking and transmission to backend.

4.6 Development Decisions and Lessons Learned

Embedding model choice matters just as much as architecture: CLIP is not suitable for text-to-text similarity. Payload consistency (image paths must match static files). API sensitivity to qdrant-client version. Separating category metadata from artwork titles. Full-text storage for section_text improves semantic precision.

5. Evaluation

This chapter evaluates the five graph generation strategies and presents findings on their comparative strengths.

5.1 Evaluation Design

Motivation for the chosen approach. The core question: which graph strategy produces the most meaningful and useful exploration paths? Description of the evaluation methodology.

5.2 Side-by-Side Strategy Comparison

Two graph modes presented simultaneously from the same starting artwork. Users or expert evaluators rate: which side produces more relevant connections; which side they would continue exploring; which side's explanations are more illuminating.

5.3 Edge Relevance Judgments

For a sample of edges from each strategy: expert rating of whether the connection is art-historically, medically, or educationally meaningful. Scale and rating criteria.

5.4 Graph-Level Statistics

Quantitative comparison across strategies: average similarity score, score variance, category distribution of connected nodes, expansion depth before stalemate. Hypothesis: text graph → higher medical-thematic coherence; image graph → more visually diverse but lower category consistency; combined graph → better balance.

5.5 Results

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6. Discussion

This chapter reflects on what the results reveal about graph-based exploration of art collections, and what remains open.

6.1 What the Comparison Reveals

Discussion of the differences between text, image, and combined strategies in practice. Why visual and semantic connections diverge for medical art. The role of LLM-generated descriptions in shaping the graph structure.

6.2 The Role of LLM-Generated Explanations

Do edge explanations improve user trust and navigation quality? Are there cases where the explanation is misleading or generic? Observations from development and testing.

6.3 Search Relevance vs. Exploration Novelty

The tension surfaced during development: the most similar artwork is not always the most valuable next step in exploration. Discussion of the exclusion mechanism and whether diversity ranking would be a better long-term approach.

6.4 Limitations

Scope: the collection of ~600 artworks is relatively small, and results may not generalize. Evaluation: preference-based comparison is subjective. The graph currently lives only in the browser session and is not persisted. LLM explanations are generated per-pair and may not be consistent across sessions.

6.5 Future Directions

Potential extensions: external knowledge enrichment (Wikidata, medical ontologies); persistent graph storage and saved sessions; user-directed exploration questions; comparison of alternative embedding models (OpenCLIP, SigLIP 2); user study with domain experts (medical historians, art historians).

7. Conclusion

Summary of contributions: a working graph-based exploration prototype for the Bagatelle medical art collection; a configurable architecture for comparing five graph generation strategies; a design for LLM-augmented edge explanation; findings on how different representation strategies shape exploratory connections. Reflection on the broader implication: that for collections where meaning is visually encoded and medically or historically specific, graph-based exploration with interpretable edges is a more appropriate interaction model than keyword search.

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